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Mechanical Relaxation Behavior of Rhyolitic Obsidian

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Accurate modelling of most volcanic or magmatic processes requires data on the rheological properties of magma [8, 14]. In this context the rheological properties of natural and synthetic silicate glasses and melts are of interest. Special investigation fields focus on the influence of chemical composition, water content, bubble and crystal content on shear modulus G*, internal friction Q^{-1} (viscoelasticity) and viscosity η [4-6a, 9, 10, 12, 13, 15].

Suggested effects on the mechanical relaxation of obsidian

	β Process	β' Process	α Process	M_{RT} Young's Modulus
Chemical Composition	X	X	X	(X)
Water Content	X	X	X	X
Crystal Content	(X)	(X)	(X)	
Bubble Content			(X)	
Temperature History	(X)	(X)	(X)	

In this study low frequency (f~0.63Hz) mechanical relaxation behavior of various rhyolitic obsidians has been investigated at 1bar in the temperature range RT-1000°C using special flexure pendulum experiments [1, 2]. Three relaxation processes are assumed: one above the glass transition temperature T_g α–relaxation (viscoelastic process) and two below T_g β–relaxation (possibly in combination with water and mixed alkali mobility) and β'-relaxation (possibly in combination with oxygen and alkaline earth mobility).

Low Frequency Flexure Pendulum Experiments

The **measurement equipment** is shown in Fig. 1 (cf. [1]). The sample is a combination of two bars with a rectangular cross section of (1x1)mm² and a lenght of 40mm. One end of the specimen is held rigid and the other end is connected with the movable part of the flexure pendulum. Measurements were done under isothermal conditions in the range below the glass transition temperature T_g in 10K-steps and above T_g in 5K-steps.

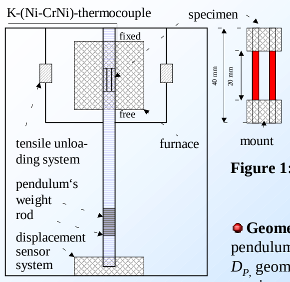

Figure 1: Flexure Pendulum Equipment

Geometrical quantities: pendulum's direction moment D_P, geometrical factors of specimen *l, I*.

Experimentally determined quantities: logarithmic decrement without specimen Λ_P and of the coupled system specimen-pendulum Λ_{SP}, oscillation period of pendulum T_P and of the coupled system specimen-pendulum T_{SP} (Fig. 2).

With the geometrical and experimentally determined quantities the rheological parameters complex Young's modulus M* and internal friction Q^{-1} were calculated [1, 2].

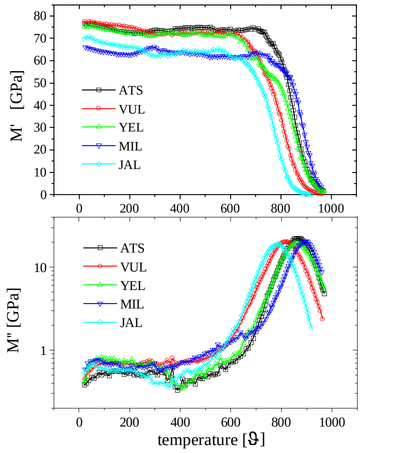

Figure 2: Temperature dependence of the real M' and imaginary part M'' of the complex Young's modulus M* for five selected natural volcanic glasses.

Sample selection

The nine natural volcanic glasses used were fresh, unweathered, unaltered, and nonhydrated obsidians free of cracks with low crystal (<1%) and bubble content (<1%).

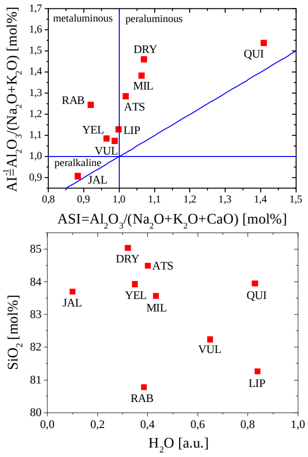

Figure 3: Chemical composition and relative water content of the investigated natural volcanic glasses.

Chemical composition of the matrix glass was quantified by electron microanalysis (SEM-EDX). Volatile-species and thus water content are determined in high-vacuum degassing experiments (controlled heating rate 10K/min from RT-1500°C) coupled with a quadropole mass spectrometer (QMA 125 Balzers) [5] (Fig. 3).

YEL is a grey crystal-poor obsidian from Yellowstone (USA). VUL from Vulcano, LIP from Lipari (Italy) and JAL from Jalisco (Mexico) are grey crystal-free glasses. The sample MIL from Milos (Greek) is grey-black crystal-poor. RAB from Hrafntinnuhryggur (Island) and DYR (Armenia) are black crystal-poor glasses. The obsidian ATS from Artenis (Armenia) is grey crystal-poor with crystal bands. QUI from Quironcolo (Argentinia) is a transparent homogeneous obsidian with very few large crystals of cordierite.

Anelastic and Viscoelastic Behavior of Rhyolitic Obsidian

Relaxation model

A **Generalized Maxwell Model** (GMM) is used to calculate anelastic and viscoelastic behavior in the restricted temperature range with one set of parameters (Fig. 4): activation energy E_i and preexponential factor $\tau_{0,i}$ of the relaxation time τ_i, moduli M_i, E_i, $X_i=M_i/E_i$, distribution parameter α_i, i=1…3 (Tab. 1). This GMM consists of three fractional Maxwell elements [11] in parallel. For α_i=0, X_i=0, $M_i\neq0$ the single fractional Maxwell elements are transformed to Maxwell elements with one relaxation time.

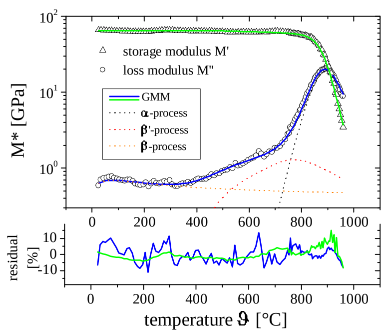

Figure 4: Temperature dependence of the real M' and imaginary part M'' of the complex Young's modulus M* for the MIL-obsidian. A Levenberg-Marquardt algorithm [7] is used to find the GMM-parameters that give the best fit.

Influences on the mechanical relaxation behavior

The individual GMM parameters are sensitive for different influences. Comparisons with other measurements (viscosity, diffusion, electrical conductivity) can be made for the activation energies E_i of the single relaxation processes (cf. Tab. 1). Below we are restricted to the α-process.

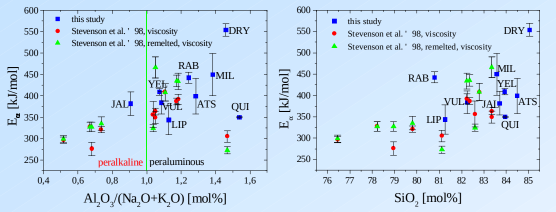

Figure 5: (left) Aktivation energy E_α as a function of Al/alkali ratio and (right) SiO_2 content for different obsidians.

Calculation of the apparent activation energy E_α'~ $\alpha_\alpha E_\alpha$ for the α-process with the arrhenius equation $f=f_0\exp\{-E_\alpha'/(RT_{max})\}$ shows the deviation from Maxwell–behavior (temperature T_{max} at the maximum of loss modulus M'', vibration frequency $f_0=k_BT_{max}/h$, k_B Boltzmann's constant, *h* Plank's constant).

Table 1: Some selected GMM-parameters for the various natural volcanic glasses.

		YEL	VUL	LIP	MIL	RAB	DYR	ATS	JAL	QUI	ERR [%]
M_α	[GPa]	31.1	15.6	6.6	13.7	1.5	2.7	12.6	18.2	7.3	< 5
E_α	[kJ/mol]	409	384	344	450	442	554	399	382	359	≈ 5 – 10
$-\log(\tau_\alpha)$	[log(s)]	18.32	17.28	15.54	19.08	18.89	24.24	17.20	18.00	15.14	< 10
α_α		0.52	0.55	0.57	0.60	0.78	0.67	0.60	0.61	0.60	≈ 5 – 15
X_α		0.51	0.23	0.16	0.23	0.03	0.04	0.20	0.34	0.10	< 20
M_{RT}	[GPa]	75	77	61	67	73	72	76	70	81	< 5
T_{max}	[K]	1136	1091	1056	1161	1102	1102	1137	1055	1084	
E'_α	[kJ/mol]	296	284	248	303	287	287	297	275	282	
$E_\alpha\alpha_\alpha$	[kJ/mol]	213	211	196	270	345	371	239	233	215	≈ 10 – 30
$E_{\beta'}$	[kJ/mol]	242	197	221	200	289	328	247	254	200	≈ 20 – 30
E_β	[kJ/mol]	30	68	32	20	69	32	80	30	52	≈ 20 – 30

Chemical composition: Al/alkali ratio and SiO_2 content have the largest influence. The activation energy E_α of viscous flow decreases with decreasing Al/alkali ratio from peraluminous to peralkalin and increases with increasing SiO_2 content (Fig. 5).

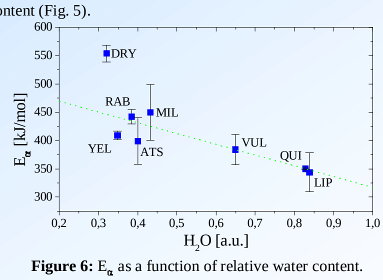

Figure 6: E_α as a function of relative water content.

Water content: The influences of water content and chemical composition cannot clearly be separated for all obsidians. However E_α decreases with increasing water content (Fig. 6) [9, 13].

Temperature history: Measurements for different temperature histories show that only a small influence on the mechanical relaxation behavior exists (for all processes).

Bubble and crystal content: The investigated natural volcanic glasses have very low bubble and crystal content (<1%). Therefore the influence is negligible [4-6a, 10, 12]. Further investigations have to show how far crystal bands have an influence on the rheological behavior and especially on the distribution parameter α_α.

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